Hospital Appointment System

Full-Stack Web Application

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Training Program	.NET Full stack Digital Engineering
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Project Type	Capstone Project

"Eliminating patient wait times through intelligent scheduling"

Problem Statement: Hospital Appointment System

Managing patient appointments in hospitals and clinics is often a chaotic and inefficient process due to the absence of a centralized scheduling system. Most healthcare facilities still rely on manual entry, phone-based bookings, or outdated software that lacks real-time availability of tracking and automated reminders.

Patients frequently experience long wait times, missed appointments due to lack of reminders, and confusion about which specialist to consult for their symptoms. On the other hand, doctors and hospital administrators struggle with scheduling conflicts, overbooking, last-minute cancellations, and the absence of consolidated availability calendars.

To address these challenges, there is a clear need for a Hospital Appointment System that provides a smart, user-friendly, and automated platform. The system must allow patients to book appointments based on real-time doctor availability, receive SMS and email reminders, and get intelligent doctor suggestions based on their reported symptoms.

Additionally, integrating a **GenAI-powered symptom-to-specialist engine** can further enhance patient experience by recommending the most appropriate doctor before the booking is even made — reducing unnecessary consultations and improving overall healthcare outcomes.

The goal of this system is to reduce patient wait times, eliminate scheduling conflicts, automate communication workflows, and digitize hospital appointment management end-to-end.

Proposed Solution: Hospital Appointment System

To overcome the inefficiencies of manual scheduling and poor patient-doctor coordination, a **full-stack web application** will be developed using Angular, .NET Web API, and SQL — automating appointment booking, availability management, and patient communication.

The system will provide three primary modules: Patient Module, Doctor/Admin Module, and GenAI Integration, supported by a secure JWT-authenticated backend.

1. Patient Module

- Secure patient registration and JWT-based login
- Search doctors by specialization, name, or symptom suggestion
- View real-time doctor availability calendar
- Book, reschedule, or cancel appointments
- Receive SMS and email reminders before appointment
- View personal appointment history and upcoming visits

2. Doctor / Admin Module

- Admin and doctor login with role-based access control
- Manage and update doctor availability calendars
- Approve, reject, or reschedule patient appointments
- Monitor daily, weekly, and monthly appointment summaries
- View patient records and consultation history
- Generate downloadable reports for hospital administration

3. Appointment Booking Engine

- Real-time slot availability checking with conflict prevention
- Date and time validation using JavaScript
- Automatic double-booking prevention at the API level
- Cancellation window enforcement (e.g., 2 hours before appointment)
- Waitlist management for fully booked doctors

4. Notifications & Reminders

- Automated SMS reminders via Twilio / SMS gateway integration
- Email reminders using SMTP / SendGrid
- Confirmation emails on booking, reschedule, and cancellation
- Reminders sent 24 hours and 1 hour before appointment

5. Dashboard & Reports

- Visual dashboard showing total bookings, cancellations, and doctor utilization
- Calendar view of all scheduled appointments per doctor
- Downloadable monthly appointment and revenue reports
- Admin analytics — peak hours, most-booked specialists, no-show rates

6. Technology-Based Implementation

Layer	Technology
Frontend (UI)	Angular 17+ (Standalone Components, Routing, Reactive Forms)
Styling	Angular Material / Bootstrap 5
Logic & Validation	TypeScript / JavaScript (date & slot validation)
Backend API	C# / ASP.NET Core Web API (.NET 8)
Authentication	JWT (JSON Web Tokens) + Role-Based Access Control
Database	SQL Server (T-SQL, Stored Procedures, Triggers)
Notifications	SMTP / SendGrid (email), Twilio (SMS)
GenAI	Anthropic Claude API / OpenAI GPT (symptom suggestion)
Testing	unit (.NET unit tests), Jasmine/Karma (Angular)
DevOps	GitHub, CI/CD pipelines, Azure deployment

7. GenAI Integration

- Symptom-based doctor recommendation engine — patient describes symptoms, AI suggests appropriate specialist
- Auto-generate daily and monthly appointment summary reports
- AI chatbot answers FAQs: "What is my next appointment?", "How do I reschedule?"
- Smart insights for admin — peak booking hours, underutilized doctors, cancellation patterns

Expected Outcomes

- Significant reduction in patient waits times through intelligent scheduling
- Elimination of scheduling conflicts via automated conflict detection
- Improved patient experience through timely SMS and email reminders
- Better doctor utilization and resource planning for hospital admin
- Transparent, data-driven appointment management and reporting
- Faster and more accurate doctor discovery through GenAI symptom matching

Project Folder Structure

Below is the industry-standard folder structure for the **Hospital Appointment System** (Angular + .NET + SQL):

```
Hospital-Appointment-System/
|
+-- frontend/                       (Angular Application)
|   +-- src/
|   |   +-- app/
|   |   |   +-- core/               # Core services (auth, API, guards)
|   |   |   |   +-- auth.service.ts
|   |   |   |   +-- api.service.ts
|   |   |   |   +-- auth.guard.ts
|   |   |   |   +-- jwt.interceptor.ts
|   |   |   |
|   |   |   +-- shared/             # Reusable components
|   |   |   |   +-- header/
|   |   |   |   +-- footer/
|   |   |   |   +-- sidebar/
|   |   |   |   +-- calendar/
|   |   |   |
|   |   |   +-- modules/
|   |   |   |   +-- auth/           # Login / Register
|   |   |   |   |   +-- login/
|   |   |   |   |   +-- register/
|   |   |   |   |
|   |   |   |   +-- patient/
|   |   |   |   |   +-- dashboard/
|   |   |   |   |   +-- book-appointment/
|   |   |   |   |   +-- appointment-history/
|   |   |   |   |   +-- symptom-checker/
|   |   |   |   |
|   |   |   |   +-- doctor/
|   |   |   |   |   +-- dashboard/
|   |   |   |   |   +-- availability/
|   |   |   |   |   +-- appointments/
|   |   |   |   |
|   |   |   |   +-- admin/
|   |   |   |   |   +-- dashboard/
|   |   |   |   |   +-- manage-doctors/
|   |   |   |   |   +-- manage-patients/
|   |   |   |   |   +-- reports/
|   |   |   |
|   |   |   +-- app-routing.module.ts
|   |   |   +-- app.component.ts
|   |   |
|   |   +-- assets/
|   |   +-- environments/
|   |
|   +-- angular.json
|   +-- package.json
|
+-- backend/                        (.NET 8 Web API)
|   +-- Controllers/
|   |   +-- AuthController.cs
|   |   +-- AppointmentController.cs
|   |   +-- DoctorController.cs
|   |   +-- PatientController.cs
|   |   +-- AdminController.cs
```

```
|   |   +-- NotificationController.cs
|   |
|   +-- Models/
|   |   +-- User.cs
|   |   +-- Appointment.cs
|   |   +-- Doctor.cs
|   |   +-- Patient.cs
|   |   +-- Notification.cs
|   |
|   +-- Services/
|   |   +-- AppointmentService.cs
|   |   +-- NotificationService.cs
|   |   +-- SymptomAIService.cs
|   |   +-- ReportService.cs
|   |
|   +-- Data/
|   |   +-- AppDbContext.cs
|   |
|   +-- DTOs/
|   |   +-- AppointmentDTO.cs
|   |   +-- DoctorDTO.cs
|   |   +-- PatientDTO.cs
|   |
|   +-- Repositories/
|   |   +-- AppointmentRepository.cs
|   |   +-- DoctorRepository.cs
|   |   +-- UserRepository.cs
|   |
|   +-- appsettings.json
|   +-- Program.cs
|
+-- database/                       (SQL Scripts)
|   +-- schema.sql                  # Tables creation
|   +-- procedures.sql              # Stored procedures
|   +-- triggers.sql                # Conflict detection triggers
|   +-- views.sql                   # Reporting views
|   +-- sample-data.sql
|
+-- tests/
|   +-- unit/
|   |   +-- AppointmentServiceTests.cs
|   |   +-- ConflictDetectionTests.cs
|   |   +-- NotificationTests.cs
|
+-- docs/
|   +-- SRS.docx
|   +-- architecture-diagram.png
|   +-- API-documentation.md
|   +-- database-ER-diagram.png
|
+-- README.md
```